

EE 445/645: Physical Models in Remote Sensing (Spring 2026)

Chapter 04-Part 01: 1D RT Problem Setup – Notes (See C4-P1 PPTs)

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Slide 1 – Title & Overview

- This opening slide introduces Chapter 4, Part 1 of the Physical Models in Remote Sensing course, focused on setting up the 1D Radiative Transfer (RT) problem. The chapter is structured to walk students through the full mathematical formulation before moving to solutions.
- The slide serves as a roadmap: it outlines the key topics to be covered — the problem's underlying assumptions, a schematic of the physical setup, the incident radiation field, the RT equation itself, a summary of input variables, and the separation of photon fields into collided and uncollided components.
- Emphasize to students that understanding the problem setup is foundational. The assumptions made here dictate the form of the equations and the scope of the model's applicability. Without a firm grasp of the setup, subsequent mathematical derivations will lack physical intuition.
- Note that the approach follows a forward-modeling framework: given known canopy and optical properties, the goal is to predict the radiation field. This stands in contrast to the inverse problem, which will also be introduced.

Slide 2 – Problem Statement: Forward and Inverse Solutions

- This slide frames the Radiative Transfer problem in terms of two complementary directions of inquiry: the forward solution and the inverse solution.
- In the forward solution, canopy structure and canopy optics are treated as known inputs. The unknown to be determined is the radiation field — both within the medium and emerging from it at the boundaries. This is the classical model-building and evaluation task: given what we know about the vegetation canopy, predict what a sensor would measure.
- In the inverse solution, the radiation field (measured by a sensor) is the known quantity, and the canopy structure and canopy optics are the unknowns to be retrieved. This is the remote sensing estimation problem — deriving physical properties of a surface from observed radiances.
- Students should appreciate that the Radiative Transfer Equation (RTE) is the mathematical bridge between both directions. Solving it forward is analytically or numerically tractable; solving it in reverse (inversion) generally requires optimization or statistical techniques. This chapter focuses on the forward problem.

Slide 3 – Assumptions of the 1D RT Model

- This slide presents the simplifying assumptions that define the 1D RT problem. Each assumption narrows the physical scenario to make the mathematics tractable while still capturing the dominant radiative processes in a vegetation canopy.
- Key assumptions include: (1) Horizontal homogeneity — the canopy properties do not vary laterally, only vertically. This is what makes the problem one-dimensional. (2) The medium consists of planar leaves only — other canopy elements like branches and stems are ignored. (3) Illumination comes from two components: direct solar radiation (unscattered by the atmosphere) and isotropic diffuse sky radiation (which has interacted with the atmosphere). (4) The problem is stationary — no time dependence. (5) No frequency shifting — photons don't change wavelength upon interaction. (6) No emission — thermal emission within the medium is neglected, limiting applicability to the solar reflective domain.
- The boundary conditions are also specified here: the top boundary is non-re-entrant (photons leaving the top don't re-enter from the top), while the bottom boundary is re-entrant (soil reflects radiation back up into the canopy).
- Students should think critically about which assumptions might break down in real-world scenarios — e.g., row crops violating horizontal homogeneity, or thermal infrared applications where emission cannot be neglected.

Slide 4 – Schematic of the 1D Problem

- This slide provides a geometric and conceptual diagram of the 1D RT problem setup. It grounds the abstract assumptions in a physically intuitive picture.
- The canopy is oriented along the z -axis, with $z = 0$ (or $L = 0$) at the top and $z = z_H$ (or $L = LAI$) at the bottom. The z -axis points upward toward the zenith, so downward-traveling photons have negative z -direction components. The leaf area index (LAI) is used as the vertical coordinate L — cumulative leaf area per unit ground area from the top of the canopy downward.
- Two illumination components are shown: direct solar radiation arriving from a specific direction Ω_0 with intensity $I_0\lambda$, and isotropic diffuse sky illumination $I_d\lambda$ arriving uniformly from the upper hemisphere.
- The notation convention is critical: directions (Ω) are measured with respect to the negative z -axis (downward). As a result, $\mu > 0$ corresponds to upward-traveling photons, and $\mu < 0$ to downward-traveling photons. Students should internalize this sign convention, as it appears throughout all subsequent equations.

Slide 5 – Incident Radiation Field

- This slide formalizes how the total incident radiation field at the top of the canopy is partitioned between its direct and diffuse components.
- The total incoming flux F_{incident} is normalized to 1 W/m^2 . It is split using the parameter f_{dir} — the fraction of incident radiation that arrives as direct solar flux. Therefore, the direct solar flux is $F_0 = f_{\text{dir}} \times F_{\text{incident}}$ and the diffuse sky flux is $F_d = (1 - f_{\text{dir}}) \times F_{\text{incident}}$.
- This decomposition matters because the RTE is linear: solutions from direct and diffuse components can be computed separately and superimposed. Linearity also means that normalizing the input to 1 W/m^2 is without loss of generality — actual irradiance values can be scaled in post-processing.
- The parameter f_{dir} is a key model input and can, in principle, be wavelength-dependent (though often it is treated as spectrally invariant). Two extreme cases: $f_{\text{dir}} = 0$ represents a completely overcast sky with only diffuse illumination, while $f_{\text{dir}} = 1$ represents a perfectly clear sky with only direct sunlight. Real conditions fall between these extremes.

Slide 6 – Upper Boundary Condition

- This slide formalizes the upper boundary condition of the 1D RT problem — i.e., what intensities are specified at the top of the canopy ($L = 0$) for downward-traveling directions ($\mu < 0$).
- The intensity at the top consists of two terms: the direct solar beam and the diffuse sky component. The direct solar component is described by a Dirac delta function in direction space — it is perfectly collimated, arriving only from the solar direction $\Omega_0 = (\theta_0, \phi_0)$, which are input angles. The diffuse sky component is characterized by an anisotropy function d_0 . For the isotropic case (uniform sky radiance), $d_0 = 1/\pi$.
- The solar zenith and azimuth angles θ_0 and ϕ_0 are required inputs. They define the geometry of the direct illumination and influence how deep solar photons penetrate the canopy before being intercepted.
- Students should note that the delta function in the direct term is what mathematically distinguishes the direct beam from the diffuse field — it creates a singularity in direction space that motivates the separation of the uncollided and collided radiation fields (discussed later).

Slide 7 – Lower Boundary Condition

- This slide specifies the lower boundary condition — the intensity of radiation entering the canopy from below (upward-traveling photons, $\mu > 0$) at the bottom of the canopy ($z = z_H$, $L = LAI$).
- The lower boundary is a reflecting surface, typically soil but possibly understory vegetation or snow. Its reflectance is described by a bi-directional reflectance distribution function (BRDF) $\rho_{s\lambda}$, which is in general wavelength-dependent and direction-dependent.
- For simplicity, an isotropic lower surface is assumed: $\rho_{s\lambda} = \rho_{g\lambda}/\pi$, where $\rho_{g\lambda}$ is the wavelength-dependent hemispherical reflectance of the ground. Under this assumption, the upwelling intensity at the bottom depends only on the total downwelling flux F_d at $z = z_H$, not on the direction of the reflected photons. The ground reflectance $\rho_{g\lambda}$ is a model input.
- This isotropic approximation is physically reasonable for many natural surfaces (dry soil, grass litter) but breaks down for wet soils, snow, or specular surfaces. More complex BRDF models could be substituted without changing the overall RT framework.

Slide 8 – The 1D Radiative Transfer Equation

- This slide presents the full 1D RTE — the governing equation of the problem. All subsequent analysis is built on solving this equation subject to the boundary conditions developed in the previous slides.
- The RTE describes how the specific intensity $I(L, \Omega)$ changes as photons travel in direction Ω through the canopy at depth L . The left-hand side represents the differential change in intensity; the right-hand side contains two competing effects: (1) extinction (attenuation due to absorption and scattering out of the beam), and (2) in-scattering (contribution from photons scattered into direction Ω from all other directions Ω').
- Although the intensity I is wavelength-dependent (subscript λ), the wavelength dependence will be dropped from the notation going forward for brevity. Students should remember, however, that the cross sections and boundary conditions are all spectral quantities.
- The boundary conditions from Slides 6 and 7 close the system. Together, the RTE plus boundary conditions form a well-posed boundary value problem whose solution yields the full radiation field throughout and at the boundaries of the canopy.

Slide 9 – The Cross Sections

- This slide defines the two fundamental cross-sectional quantities that appear in the RTE: the extinction cross section and the scattering phase function.
- If the leaf normal orientation distribution function g_L is assumed depth-independent (i.e., the same throughout the canopy), the cross sections simplify considerably. The extinction cross section σ_e involves the geometry factor $G(\Omega)$, which describes the mean projected area of leaves in direction Ω per unit leaf area. G depends on the leaf normal distribution and can be computed analytically for common distributions (e.g., spherical, planophile, erectophile).
- The area scattering phase function $\Gamma(\Omega' \rightarrow \Omega)$ describes the probability of a photon traveling in direction Ω' being scattered into direction Ω . It is computed as an integral involving the leaf scattering phase function γ_L , which captures the optical properties (reflectance and transmittance) of individual leaves.
- The leaf area density $u_L(z)$ describes the vertical distribution of leaf area in the canopy. When multiplied by the appropriate cross sections, it gives the volumetric scattering and extinction properties at each depth. These derivations are detailed and students are encouraged to trace through the definitions carefully.

Slide 10 – Switch from z to L (Cumulative LAI Coordinate)

- This slide introduces a coordinate transformation that simplifies the RT equations: replacing the geometric depth z with the cumulative leaf area index L .
- The transformation is achieved by dividing the RTE through by the leaf area density u_L . This change of variable is natural because leaf area index — not geometric thickness — is what determines the probability of a photon intercepting a leaf. Two canopies of different geometric depths but equal LAI will attenuate radiation identically (given the same G factor).
- After this transformation, the canopy occupies the domain $L \in [0, L_H]$, where $L_H = \text{LAI}$ is the total leaf area index of the canopy (a model input). The top of the canopy is at $L = 0$ and the bottom at $L = L_H$.
- This formulation is common in vegetation RT models and connects directly to field-measurable quantities. LAI is routinely retrieved from ground measurements, optical instruments (e.g., LAI-2200), or remote sensing data — making L a practical and physically meaningful vertical coordinate.

Slide 11 – Summary of Input Variables

- This slide provides a consolidated reference of all required model inputs, organized into four categories. It is useful to return to this slide as a checklist when implementing or running the RT model.
- Category 1 — Incident radiation field: The input f_{dir} specifies the fraction of direct solar flux in the total incident radiation. It may be wavelength-dependent.
- Category 2 — Canopy structure (wavelength-independent): The leaf area index L_H (= LAI) defines the canopy depth. The leaf normal orientation distribution $g_L(\Omega_L)$ is specified through a model for $g_L(\theta_L)$ (e.g., spherical or planophile distribution), with uniform azimuthal distribution $h_L = 1$ assumed.
- Category 3 — Canopy optics (wavelength-dependent): The leaf scattering phase function γ_L is used, with specular reflection neglected. A bi-Lambertian model is adopted for diffuse scattering, and the leaf reflectance ρ_{Ld} and transmittance τ_{Ld} are inputs, assumed identical for both leaf surfaces.
- Category 4 — Ground optics (wavelength-dependent): The ground reflectance $\rho_{g\lambda}$ is an input, with the lower surface assumed to be a Lambertian (isotropic) reflector.
- Students should note the distinction between wavelength-dependent and wavelength-independent inputs — this structure is important for understanding how the model is run spectrally.

Slide 12 – Separation into Collided and Uncollided Problems

- This slide introduces one of the most important analytical tools in 1D RT theory: the decomposition of the total radiation field into uncollided and collided components.
- The total specific intensity I is written as $I = I_0 + I_C$, where I_0 is the intensity of photons that have not yet interacted with any leaf element (uncollided), and I_C is the intensity of photons that have experienced at least one collision (collided).
- This separation is motivated by the singularity introduced by the direct solar beam in the upper boundary condition. The delta function in direction space (representing the collimated solar beam) cannot be handled by standard numerical methods for the full equation. By peeling off the uncollided component, the singularity is isolated and solved analytically, leaving a smoother, more numerically tractable equation for the collided field.
- The decomposition transforms the single boundary value problem into two coupled sub-problems: an uncollided problem (solved first, analytically) and a collided problem (solved second, using the uncollided solution as a source term). This approach is standard in nuclear transport theory and has been successfully applied to vegetation canopy RT modeling.

Slide 13 – The Uncollided Problem

- This slide presents the RTE specifically for the uncollided radiation field I_0 . Because uncollided photons, by definition, have not scattered, there is no in-scattering source term — the equation contains only the extinction term.
- This simplification makes the uncollided problem a first-order linear ODE in L (for a fixed direction Ω), which can be solved analytically. There is no integral (scattering) source term to contend with.
- The boundary conditions for the uncollided problem are the same as those at the top of the canopy for downward directions: the direct solar beam and the diffuse sky radiation. Importantly, only photons entering from outside (through the upper or lower boundaries) can be uncollided — once inside the canopy, any interaction produces a collided photon.
- The analytical tractability of the uncollided problem is a key advantage of this decomposition strategy. The solution, derived on the next slides, takes the form of an exponential attenuation — physically, the Beer-Lambert law generalized to anisotropic media.

Slide 14 – The Collided Problem

- This slide presents the RTE for the collided radiation field I_C . Unlike the uncollided problem, the collided equation includes a source term that drives the generation of collided photons.
- The total source on the right-hand side has two components: (1) Q , the first-collision source — this accounts for photons from the uncollided field I_0 that undergo their first scattering event and thereby join the collided population. (2) S , the distributed (multiple-collision) source — this accounts for photons already in the collided field I_C being scattered again, contributing further to I_C .
- The collided problem is more complex than the uncollided problem because it is an integro-differential equation (the source S involves an integral of I_C over all directions). Standard numerical methods — such as discrete ordinates, spherical harmonics, or Monte Carlo — are typically applied to solve it.
- The first-collision source Q is computed using the already-known uncollided solution I_0 , making the two sub-problems sequentially solvable: solve the uncollided problem first, then use its solution to construct Q and solve the collided problem.

Slide 15 – Solution of the Uncollided Problem

- This slide gives the analytical solution to the uncollided problem for both downward and upward directions.
- For downward directions ($\mu < 0$): the uncollided intensity at depth L in direction Ω is the incident intensity at the top boundary ($L = 0$) multiplied by the transmission probability $T(0 \rightarrow L, \Omega)$ — the probability that a photon traveling in direction Ω from $L = 0$ reaches depth L without any collision. This transmission function has the form of a generalized Beer-Lambert exponential.
- For upward directions ($\mu > 0$): the uncollided intensity at depth L is the intensity entering from the lower boundary (at $L = L_H$) multiplied by the transmission probability $T(L_H \rightarrow L, \Omega)$ — the probability of traveling from the bottom upward to depth L without collision.
- The transmission probability $T(L_1 \rightarrow L_2, \Omega)$ generalizes the standard Beer-Lambert law to the case of an anisotropic medium (where the extinction depends on direction through the geometry factor $G(\Omega)/|\mu|$). It evaluates to $\exp(-G(\Omega)/|\mu| \times |L_2 - L_1|)$ for a homogeneous canopy. These solutions are exact within the model assumptions and require no numerical approximation.

Slide 16 – Pictorial Illustration: Uncollided Solution (Downward)

- This slide provides a visual explanation of the downward uncollided solution to build physical intuition.
- The diagram illustrates a photon traveling downward in direction Ω from the top of the canopy ($L = 0$) to depth L . The uncollided intensity at depth L is simply the incident intensity at the top, attenuated by the probability that the photon has not been intercepted by a leaf during its journey.
- The transmission probability — the exponential factor — represents the fraction of photons that 'survive' from $L = 0$ to L without collision. This probability depends on the direction Ω (through the geometry factor G and the cosine μ of the zenith angle): more oblique directions (smaller $|\mu|$) pass through more leaf material per unit depth, leading to stronger attenuation.
- Students should connect this expression to the direct transmittance of the canopy for solar radiation — often denoted P_{gap} or τ_{direct} in the literature. The uncollided solar beam intensity at depth L is precisely this gap probability times the incident intensity, making the uncollided solution directly relatable to measurable quantities like sunfleck statistics.

Slide 17 – Pictorial Illustration: Uncollided Solution (Upward)

- This final slide provides the analogous visual explanation for the upward uncollided solution.
- Here, a photon is traveling upward in direction Ω from the lower boundary ($L = L_H$) toward depth L within the canopy. The uncollided upward intensity at depth L is the intensity entering from the lower boundary (determined by the lower boundary condition — soil reflectance times the downwelling flux), attenuated by the probability of traveling from L_H up to L without a collision.
- The structure of the solution is identical to the downward case, but the reference boundary is now the bottom ($L = L_H$) rather than the top. The transmission factor captures the attenuation as the reflected photon propagates upward through the canopy.
- These two solutions — downward and upward uncollided fields — together provide the complete first-collision source Q needed to solve the collided problem (Slide 14). Having derived these analytical expressions, the next part of the course will address numerical methods for solving the full coupled problem and extracting observable quantities such as canopy reflectance, transmittance, and absorbed radiation.

Uncollided 1D Radiative Transfer

Figure Highlights — RED and NIR Bands

Model Setup & Inputs

Both figures share the same geometry and canopy structure. The only quantities that differ between them are the leaf and ground optical properties.

- **Incident radiation:** Total flux $F_{in} = 1 \text{ W m}^{-2}$, split as $f_{dir} = 0.7$ (direct solar) and $1 - f_{dir} = 0.3$ (isotropic diffuse sky).
- **Solar geometry:** $\theta_0 = 140^\circ$ from the upward zenith axis, giving $\mu_0 = \cos(140^\circ) \approx -0.766$ (a downward direction). Azimuth $\varphi_0 = 0^\circ$.
- **Canopy structure:** Spherical leaf normal distribution $\Rightarrow G = 0.5$ for all directions (geometry factor is isotropic). $LAI = 1.5$ and $LAI = 4.0$.
- **RED band optics:** $\rho_L = 0.06$, $\tau_L = 0.04$, $\omega_L = 0.10$ (low-albedo, strongly absorbing leaves); $\rho_g = 0.10$.
- **NIR band optics:** $\rho_L = 0.525$, $\tau_L = 0.45$, $\omega_L = 0.975$ (high-albedo, near-transparent leaves); $\rho_g = 0.20$.

Figure 1 — RED Band

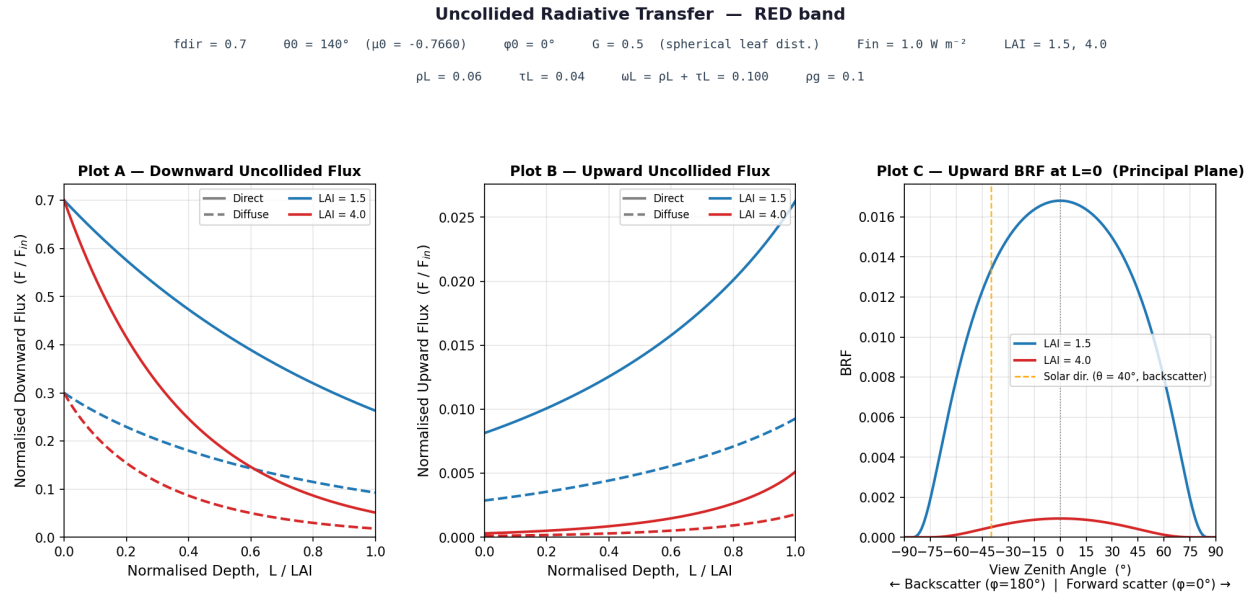


Figure 1. Uncollided radiative transfer — RED band ($\omega_L = 0.10$, $p_g = 0.10$). Left: normalised downward flux. Centre: normalised upward flux. Right: upward BRF at canopy top in the principal plane.

Plot A — Downward Uncollided Flux (RED)

- At $L/LAI = 0$ (canopy top), the direct flux starts at 0.70 and the diffuse flux at 0.30, correctly partitioning the unit incident flux.
- Both components decay monotonically with depth following a Beer–Lambert exponential $\exp(-G \cdot L / |\mu|)$. The direct beam decays faster because it travels at a fixed oblique angle ($\mu_0 \approx -0.766$); the diffuse field has contributions from all downward directions, including near-vertical ones that penetrate more efficiently, so its effective decay is slower.
- The two LAI curves overlay exactly on the normalised axis — because the shape of the attenuation profile is determined only by μ_0 and G , not by LAI itself. LAI sets the physical depth of the canopy, but the normalised profile is identical for both.
- By $L/LAI = 1.0$ (ground), the direct flux has decayed to $\approx 27\%$ and the diffuse to $\approx 10\%$ of their respective top-of-canopy values, reflecting the low-sun geometry (oblique beam path).

Plot B — Upward Uncollided Flux (RED)

- The upward flux is driven entirely by isotropic ground reflection: the ground absorbs the downward uncollided flux and reflects it isotropically with $p_g = 0.10$. Because RED leaves are strongly absorbing, only ground-reflected photons contribute to the uncollided upward field — there is no leaf-scattered component in the uncollided problem.
- The upward flux is largest at the ground ($L/LAI = 1$) and decays toward the top as photons are extinguished while travelling back up through the canopy.
- LAI = 1.5 (blue) produces significantly higher upward fluxes than LAI = 4.0 (red) at every depth, because more downward flux reaches the ground in a shallower canopy (higher F_d at $L = LAI$), and more of that reflected flux escapes back upward through less canopy.

- The upward fluxes are an order of magnitude smaller than the downward fluxes, reflecting the combined effects of canopy extinction going down, ground reflectance (10%), and canopy extinction again going back up.

Plot C — Upward BRF at Canopy Top, Principal Plane (RED)

- The BRF is computed at $L = 0$ for view zenith angles $0\text{--}90^\circ$ in both the backscatter ($\varphi = 180^\circ$, left side) and forward-scatter ($\varphi = 0^\circ$, right side) directions.
- Because the ground is isotropic and the upward uncollided intensity from below depends only on $|\mu_{\text{view}}|$ (not azimuth), the BRF is symmetric about nadir and identical on both sides of the principal plane. There is no hot spot in the uncollided field — that feature arises only from leaf single-scattering in the collided solution.
- The BRF peaks near nadir (view zenith $\approx 0^\circ$) and falls to zero at 90° , because the canopy transmission factor $\exp(-G \cdot \text{LAI} / \mu_{\text{view}})$ goes to zero as $\mu_{\text{view}} \rightarrow 0$ (grazing view through the full canopy depth).
- $\text{LAI} = 4.0$ yields a BRF nearly indistinguishable from zero: the deep canopy extinguishes essentially all ground-reflected photons before they reach the top. $\text{LAI} = 1.5$ produces a small but non-trivial BRF peak of ≈ 0.017 .
- RED BRF values are small overall (≤ 0.017) because $\rho_g = 0.10$ and the canopy attenuates the signal both ways.

Figure 2 — NIR Band

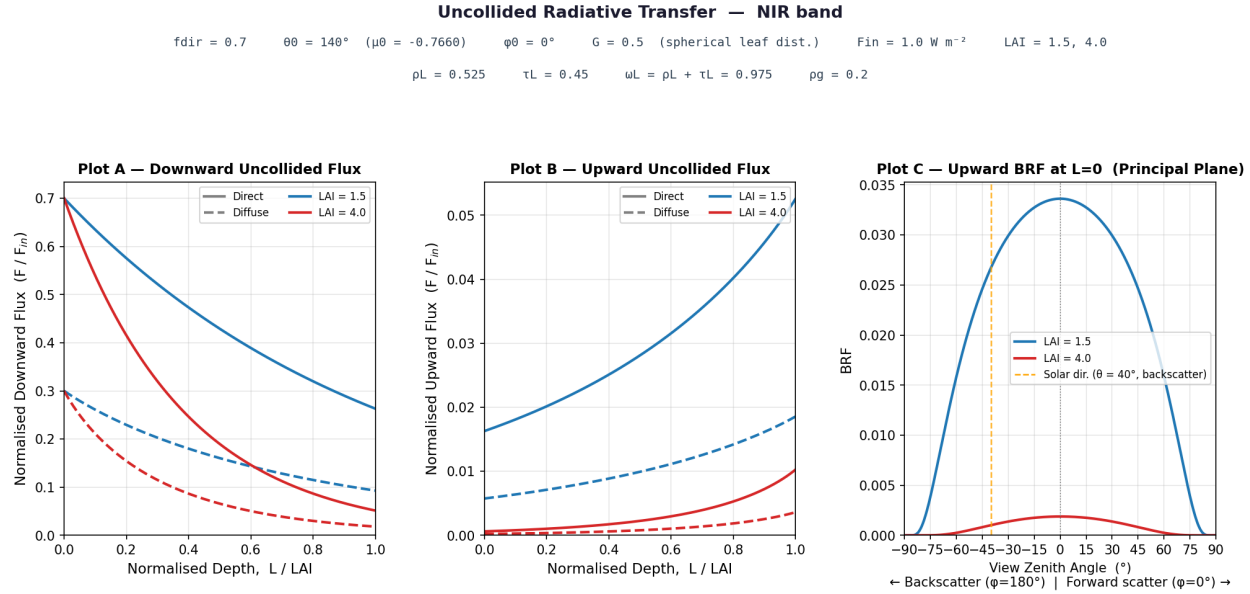


Figure 2. Uncollided radiative transfer — NIR band ($\omega_L = 0.975$, $p_g = 0.20$). Left: normalised downward flux. Centre: normalised upward flux. Right: upward BRF at canopy top in the principal plane.

Plot A — Downward Uncollided Flux (NIR)

- The downward flux profiles are identical to the RED band. This is because the uncollided downward field depends only on geometry (G , μ_0) and the incident flux — not on leaf optical properties. Leaf albedo (ω_L) and ground reflectance (p_g) affect only the collided and upward fields.
- This is a key physical result: the uncollided downward flux is wavelength-independent, determined entirely by canopy architecture and illumination geometry.

Plot B — Upward Uncollided Flux (NIR)

- The NIR upward fluxes are approximately twice those of the RED band at every depth, directly reflecting the doubled ground reflectance ($p_g = 0.20$ vs 0.10). Because the ground is the sole source of upward uncollided photons and acts as a linear reflector, the upward field scales exactly with p_g .
- For LAI = 1.5, the direct-beam upward flux (solid blue) reaches ≈ 0.055 at the ground and decays to ≈ 0.026 at the top, compared to ≈ 0.028 and ≈ 0.013 in the RED. The ratio is exactly 2.0 throughout.
- LAI = 4.0 again shows strongly suppressed upward flux relative to LAI = 1.5, for the same reasons as in the RED band: less flux reaches the deep ground, and what is reflected is more attenuated on the way back up.

Plot C — Upward BRF at Canopy Top, Principal Plane (NIR)

- The NIR BRF shape is identical to the RED — symmetric, peaking at nadir, zero at the horizon — but scaled upward by a factor of ≈ 2 due to the higher ground reflectance ($p_g = 0.20$ vs 0.10). The LAI = 1.5 peak reaches ≈ 0.034 , double the RED value.

- Again, there is no azimuthal dependence in the uncollided BRF because the Lambertian ground radiates isotropically and the uncollided upward transmission $\exp(-G \cdot \text{LAI} / \mu_{\text{view}})$ is direction-symmetric about nadir.
- The absence of a hotspot and the $\cos(\theta_{\text{view}})$ -driven roll-off toward the horizon are hallmarks of the uncollided solution. The collided field will introduce single-scattering from leaves, which will add angular structure, spectral contrast, and a backscatter peak in the principal plane.

Cross-Band Comparison

The two figures together illustrate a number of physically important contrasts between the RED and NIR spectral bands in the uncollided problem:

- **Downward flux is wavelength-independent:** Plots A are identical in both figures. The uncollided downward field is a pure geometric result — a testament to the fact that photon interception probability depends on canopy structure, not leaf colour.
 - **Upward flux and BRF scale linearly with ρ_g :** The NIR upward values are uniformly double those of RED at every depth and view angle, because $\rho_g(\text{NIR}) / \rho_g(\text{RED}) = 2.0$. In the uncollided problem, the ground is the only source of upward photons.
 - **LAI effect is identical in both bands:** The relative suppression of flux and BRF between $\text{LAI} = 1.5$ and $\text{LAI} = 4.0$ is the same in both bands, again because the attenuation is purely geometric.
 - **Spectral contrast is absent in downward flux:** The large difference in leaf albedo ($\omega_L = 0.10$ vs 0.975) has no effect on the downward uncollided field. Spectral contrast in reflectance — the basis of vegetation remote sensing indices like NDVI — emerges from leaf scattering, which is a collided-field effect.
 - **The uncollided BRF is small:** Even for the optically bright NIR band with $\text{LAI} = 1.5$, the peak BRF is only ≈ 0.034 . The bulk of the canopy reflectance signal comes from collided (scattered) photons, highlighting why the collided solution is essential for realistic canopy modelling.
-